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<https://swayam.gov.in>https://swayam.gov.in/nc_details/NPTEL

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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction To Machine Learning - IITKGP (course)



Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

- ☐ Lecture 18: Bayesian Learning (unit? unit=32&lesson=33)

- ☐ Lecture 19: Naive Bayes (unit?

Week 4 : Assignment 4

The due date for submitting this assignment has passed.

Due on 2024-08-21, 23:59 IST.

Assignment submitted on 2024-08-21, 19:02 IST

1)

2 points

A man is known to speak the truth 2 out of 3 times. He throws a die and reports that the number obtained is 4. Find the probability that the number obtained is actually 4 :

- A. $2/3$
B. $3/4$
C. $5/22$
D. $2/7$

☐ A.

☐ B.

☐ C.

☒ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

D.

2)

2 points

Two cards are drawn at random from a deck of 52 cards without replacement. What is the probability of drawing a 2 and an Ace in that order?

- A. $4/51$
B. $1/13$
C. $4/256$
D. $4/663$

☐ A.

unit=32&less
n=34)

☐ Lecture 20:
Bayesian
Network (unit?
unit=32&less
n=35)

☐ Lecture 21:
Python
Exercise on
Naive Bayes
(unit?
unit=32&less
n=36)

☐ Lecture 22:
Tutorial 4
(unit?
unit=32&less
n=37)

☐ Week 4 :
Lecture
Material (unit?
unit=32&less
n=38)

☒ **Quiz: Week 4
: Assignment
4
(assessment?
name=145)**

☐ Feedback
Form for Week
4 (unit?
unit=32&less
n=123)

☐ Assignment 4
Solution (unit?
unit=32&less
n=155)

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

**Assignment
Solution ()**

- ☐ B.
☐ C.
☒ D.

Yes, the answer is correct.

Score: 2

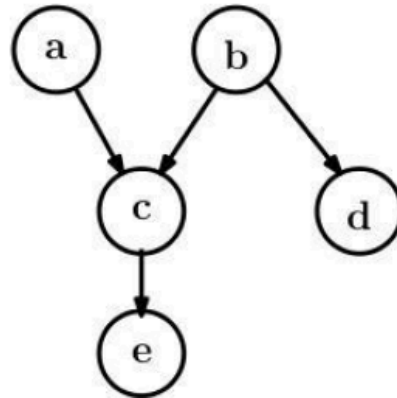
Accepted Answers:

D.

3)

2 points

Consider the following graphical model, mark which of the following pair of random variables are independent given no evidence?



- A. a,b
B. c,d
C. e,d
D. c,e

- ☒ A.
☐ B.
☐ C.
☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

A.

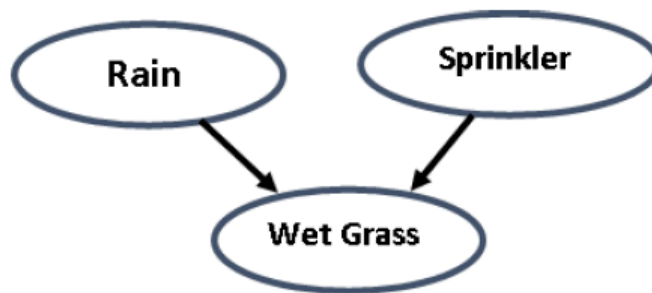
4)

2 points

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Videos ()

Problem
Solving
Session -
July 2024 ()

Consider the following Bayesian network. The random variables given in the model are modeled as discrete variables (Rain = R, Sprinkler = S and Wet Grass = W) and the corresponding probability values are given below. (Note: ($\neg$ X) represents complement of X)



$P(R) = 0.1$
 $P(S) = 0.2$
 $P(W | R, S) = 0.8$
 $P(W | R, \neg S) = 0.7$
 $P(W | \neg R, S) = 0.6$
 $P(W | \neg R, \neg S) = 0.5$

Calculate $P(S | W, R)$.

- A. 1
- B. 0.5
- C. 0.22
- D. 0.78

- ☐ A.
☐ B.
☒ C.
☐ D.

Yes, the answer is correct.
Score: 2

Accepted Answers:
C.

5)

2 points

What is the naive assumption in a Naive Bayes Classifier?

- A. All the classes are independent of each other
- B. All the features of a class are independent of each other
- C. The most probable feature for a class is the most important feature to be considered for classification
- D. All the features of a class are conditionally dependent on each other.

- ☐ A.
☒ B.
☐ C.
☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

B.

6)

2 points

A drug test (random variable T) has 1% false positives (i.e., 1% of those not taking drugs show positive in the test), and 5% false negatives (i.e., 5% of those taking drugs test negative). Suppose that 2% of those tested are taking drugs. Determine the probability that somebody who tests positive is actually taking drugs (random variable D).

- A. 0.66
- B. 0.34
- C. 0.50
- D. 0.91

- ☒ A.
- ☐ B.
- ☐ C.
- ☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

A.

7)

2 points

It is given that $P(A|B) = 2/3$ and $P(A|\bar{B}) = 1/4$. Compute the value of $P(B|A)$.

- A. $\frac{1}{2}$
- B. $\frac{2}{3}$
- C. $\frac{3}{4}$
- D. Not enough information.

- ☐ A.
- ☐ B.
- ☐ C.
- ☒ D.

Yes, the answer is correct.

Score: 2

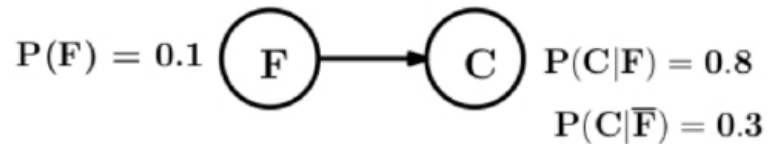
Accepted Answers:

D.

8)

2 points

Consider the following Bayesian network, where F = having the flu and C = coughing:



Find $P(C)$ and $P(F|C)$.

- A. 0.35, 0.23
- B. 0.35, 0.77
- C. 0.24, 0.024
- D. 0.5, 0.23

- ☐ A.
- ☒ B.
- ☐ C.
- ☐ D.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A.

9)

2 points

Bag I contains 4 white and 6 black balls while another Bag II contains 4 white and 3 black balls. One ball is drawn at random from one of the bags and it is found to be black. Find the probability that it was drawn from Bag I.

- A. 1/2
- B. 2/3
- C. 7/12
- D. 9/23

- ☐ A.
- ☐ B.
- ☒ C.
- ☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

C.

10)

2 points

In a Bayesian network a node with only outgoing edge(s) represents

- A. a variable conditionally independent of the other variables.
- B. a variable dependent on its siblings.
- C. a variable whose dependency is uncertain.
- D. None of the above.

- ☒ A.
- ☐ B.

☐ C.

☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

A.

